

UNIVERSITY OF OXFORD
SOFTWARE ENGINEERING PROGRAMME

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Part-time postgraduate study in software engineering



Computer Vision, VIS

10th – 14th November 2025
ASSIGNMENT

The purpose of this assignment is to test the extent to which you have achieved the learning objectives of the course. As such, your answer must be substantially your own original work. Where material has been quoted, reproduced, or co-authored, you should take care to identify the extent of that material, and the source or co-author.

Your answers to the questions on this assignment should be submitted using the Software Engineering Programme website — www.softeng.ox.ac.uk — following the submission guidelines. When submitting the assignment online, it is important that you formally complete all three assignment submission steps: step 1, read through the declaration; step 2, upload your files; step 3, check your files. **Please ensure your submission is anonymous: do not include your name or any other identifying information on the assignment, nor in accompanying material such as source code, nor within the file names of anything submitted.**

The deadline for submission is 12 noon on Tuesday, 6th January 2026. You are strongly encouraged to submit a version well before the deadline. You may update your submission as often as you like before the deadline, but no submissions or changes will be accepted after the deadline.

We hope to have preliminary results and comments available during the week commencing Monday, 16th February 2026. The final results and comments will be available after the subsequent examiners' meeting. Exam Conventions can be found here <https://www.softeng.ox.ac.uk/handbook/>

**ANY QUERIES OR REQUESTS FOR CLARIFICATION
REGARDING THIS ASSIGNMENT OR PROBLEMS INSTALLING
SOFTWARE SHOULD, IN THE FIRST INSTANCE, BE DIRECTED
TO THE PROGRAMME OFFICE WITHIN THE NEXT TWO
WEEKS.**

VIS: Assignment

Class Iteration: Week of 10-17 November 2025

Ronald Clark, PhD

This assignment will broadly test your understanding of the learning material taught within this course. Specifically, the assignment will test how well you meet the course outcomes below. After the conclusion of the VIS course, students should:

- Have a good understanding of the principles of computer vision
- Be able to apply the fundamentals to solve common tasks
- Be able to reason about an unseen problems and present a solution based on computer vision techniques, clearly discussing its strengths and weaknesses

Your answers will not only be marked on whether the final solution works, they will be scored according to:

1. the **extent to which your report shows understanding** of the course material
2. the **clarity and presentation** of your solution
3. the **technical precision and depth** of your solution

Please submit a report of **no more than 8 pages following the NeurIPS template** (see below) describing your solution. Please include diagrams or plots to aid in the explanation of your method and evaluation.

You are free to consult any reference material (eg. course notes, Google, ChatGPT, WolframAlpha) in completing this assignment, however, all answers should be original and your own work. If you consult any material that is not your own, cite the source and make clear the extent of any derivative material.

Please structure your report so that there is a cover sheet which contains the subject and date, and then attach your 8-page report following the NeurIPS template.

Submit a single PDF file containing your report and a supplementary ZIP archive with code, instructions, and necessary scripts.

1 Overview

Tracking small objects in video sequences is a fundamental yet challenging problem in computer vision, particularly when the objects are fast moving and only a few pixels in size. Traditional tracking algorithms often fail when the target object occupies a minimal number of pixels, leading to drift or complete tracking failure. Birds in flight present an example of this challenge, as they move rapidly against cluttered backgrounds and vary in scale and orientation. Effective bird tracking has applications to conservation efforts, agricultural monitoring, and the safe deployment of wind farms.

2 Problem description

This assignment focuses on developing a compute-efficient method for small object tracking, with a specific application to tracking birds. You are required to design, implement, and evaluate a novel tracking approach under realistic compute constraints. The goal is to demonstrate a well-reasoned solution, not necessarily to invent an algorithm from first principles.

A key requirement in this assignment is you may not train or adapt weights of any deep learning models. You may use off-the-shelf models (e.g. CLIP, SAM, RAFT, DINOv2) without any finetuning. Hand-crafted CV (optical flow, background subtraction, keypoint matching, homography stabilization) is allowed.

2.1 Dataset

You will use the SMOT4SB dataset [1], aimed at small multi object tracking for spotting birds. The **SMOT4SB** dataset is available at:

https://drive.google.com/drive/folders/1Y1J13W6VlgDh-L28n_mVbs7HIfo_Hv5s It contains annotated video sequences of flying birds under diverse environmental conditions. Each sequence is accompanied by ground-truth bounding boxes. Students should familiarize themselves with the data structure and preprocessing requirements before algorithm development. The dataset contains a `train` split (which can be used for developing and tuning your method) and a `pub_test` split for validation.

2.2 Baseline Method

You are required to decide on a suitable baseline approach (e.g. a pretrained YOLO model) to benchmark your approach against.

At minimum, your method must be compared against this baseline in terms of tracking accuracy (eg. precision, recall, IoU, SO-HOTA, DotD) and computational efficiency (e.g.,

runtime per frame, memory usage). For higher marks, additional comparisons against recent state-of-the-art small object trackers are encouraged.

2.3 Compute Constraints

All models must be performed under realistic compute constraints. All experiments must be carried out on CPU. Methods requiring extensive computational resources may be penalized.

3 Report Format

Your method and results should be written up in a report which contains the following sections:

- **Introduction:** Description of the problem and your approach to it.
- **Method:** A detailed overview of your approach including an overview diagram.
- **Experiments and discussion:** Comparison of your approach to the baseline (and possibly against the state of the art), including a discussion on the results
- **Conclusion:** Future work and summary.

Your report should be formatted as an 8-page NeurIPS-style paper. Use the official style files:

- NeurIPS style template:
<https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip>
- Overleaf template (NeurIPS 2024):
<https://www.overleaf.com/latex/templates/neurips-2024/tpsbbbrdqcmsh>

Not adhering to the formatting or page length requirement will result in penalties. Ensure that figures, tables, and references fit within the page limit. The 8 pages **does not** include references.

References

- [1] MVA2025 Organizing Committee. Small multi-object tracking for spotting birds (smot4sb) challenge, 2025. URL: <https://mva-org.jp/mva2025/index.php?id=challenge#footnote-1>.